

Expt 7	Spectral Estimation
Date:	7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

Code:

```

clc;
clear;
close all;

%% Signal Parameters
fs = 1000;           % Sampling Frequency
N = 256;             % Number of samples
t = (0:N-1)/fs;

% Test Signal (100 Hz and 110 Hz)
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);

%% FFT Parameters
NFFT = 2048;
f = (0:NFFT/2-1)*(fs/NFFT);

%% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hann'};

%% Apply Windows and FFT
P = cell(1,3);

for i = 1:3

    xw = x(:).*w{i};

    X = fft(xw,NFFT);

    P{i} = abs(X(1:NFFT/2));

    % Normalize
    P{i} = P{i}/max(P{i});

end

%% Individual Spectra
figure;

for i = 1:3

    subplot(3,1,i)

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    plot(f,20*log10(P{i}+eps),'LineWidth',1.5)

    title(['Spectrum using ' names {i} ' Window'])

    xlabel('Frequency (Hz)')

    ylabel('Magnitude (dB)')

    xlim([0 250])

    ylim([-100 5])

    grid on

end

%% Comparison Plot
figure

plot(f,20*log10(P{1}+eps),'k','LineWidth',1.5)
hold on
plot(f,20*log10(P{2}+eps),'r','LineWidth',1.5)
plot(f,20*log10(P{3}+eps),'b','LineWidth',1.5)

legend(names,'Location','best')

xlabel('Frequency (Hz)')

ylabel('Magnitude (dB)')

title('Comparison of Window Functions')

xlim([50 170])

ylim([-100 5])

grid on

%% Window Shapes
figure

for i = 1:3

    subplot(3,1,i)

    plot(w{i},'LineWidth',1.5)

    title([names {i} ' Window'])

    xlabel('Samples')

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        ylabel('Amplitude')

        grid on

    end

    %% Peak Detection
    pks = cell(1,3);
    loc = cell(1,3);

    fprintf('\nDetected Spectral Peaks\n');
    fprintf('-----\n');

    for i = 1:3

        [pks{i},idx] = findpeaks(P{i}, ...
            'MinPeakHeight',0.3,...
            'MinPeakDistance',10);

        loc{i} = f(idx);

        fprintf('\n%s Window:\n',names{i});

        disp(loc{i})

    end

    %% Peak Marking (Hamming Window)
    figure

    plot(f,20*log10(P{2}+eps),'LineWidth',1.5)

    hold on

    plot(loc{2},20*log10(pks{2}+eps),'ro',...
        'MarkerFaceColor','r','MarkerSize',7)

    title('Detected Peaks - Hamming Window')

    xlabel('Frequency (Hz)')

    ylabel('Magnitude (dB)')

    xlim([50 170])

    ylim([-100 5])

    grid on

```

```

%% Two Highest Peaks
fprintf('\nMain Spectral Peaks\n');
fprintf('-----\n');

for i = 1:3

    [sortedPeaks,index] = sort(P{i},'descend');

    peakFreq = [];

    k = 1;

    while length(peakFreq) < 2

        f0 = f(index(k));

        if isempty(peakFreq) || min(abs(peakFreq-f0)) > 5
            peakFreq(end+1) = f0;
        end

        k = k + 1;

    end

    fprintf('%s Window : %.2f Hz and %.2f Hz\n', ...
        names{i},peakFreq(1),peakFreq(2));

End

```

Outputs:

Command Window

Main Spectral Peaks

Rectangular Window : 100.10 Hz and 109.86 Hz

Hamming Window : 100.10 Hz and 109.86 Hz

Hann Window : 100.10 Hz and 109.86 Hz

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